Locally polynomial Hilbertian additive regression

Kim Donghyun
Dept. of Statistics, SNU

September 12, 2025

Additive model with Hilbertian response

Let $\mathbf{X} \in [0,1]^d$ be the covariate vector and $Y \in \mathbb{H}$ with separable Hilbert space $\mathbb{H}$ equipped with addition $\oplus$ and scalar multiplication $\odot$.

$$\mathbf{Y}_i = \mathbf{m}_0 \oplus \mathbf{m}_1(X_{i1}) \oplus \cdots \oplus \mathbf{m}_d(X_{id}) \oplus \epsilon_i$$

with $E(\epsilon) = \mathbf{0} \in \mathbb{H}$.

$$E(\mathbf{Y}|\mathbf{X} = \mathbf{x}) = \mathbf{m}_0 \oplus \mathbf{m}_1(x_1) \oplus \cdots \oplus \mathbf{m}_d(x_d)$$

Spaces of Hilbertian maps

- $\mathcal{M}$ be the space of tuples of $rd+1$ Hilbertian maps $\mathbf{f}^{F,\mathrm{tp}} = (\mathbf{f}, \mathbf{f}_{1,1}, \ldots, \mathbf{f}_{r,1}, \ldots, \mathbf{f}_{1,d}, \ldots, \mathbf{f}_{r,d})^\top$ with $\mathbf{f}, \mathbf{f}_{l,j} : [0,1]^d \to \mathbb{H}$ satisfying $\mathbf{f}, \mathbf{f}_{l,j} \in L_2\left(([0,1]^d, [0,1]^d \cap \mathcal{B}(\mathbb{R}^d), P\mathbf{X}^{-1}), \mathbb{H}\right)$ which means $\int_{[0,1]^d} \|\mathbf{f}(\mathbf{x})\|^2 p(\mathbf{x}) d\mathbf{x} < \infty$ and $\int_{[0,1]^d} \|\mathbf{f}_{l,j}(\mathbf{x})\|^2 p(\mathbf{x}) d\mathbf{x} < \infty$.
- $\mathcal{M}_j = \{\mathbf{f}^{F,\mathrm{tp}} \in \mathcal{M} : \mathbf{f}_{l,k} = \mathbf{0}$ for all $1 \leq l \leq r$ and $1 \leq k \neq j \leq d$ with $\mathbf{f}(\mathbf{x}) = \mathbf{f}_j(x_j)$ and $\mathbf{f}_{l,j}(\mathbf{x}) = \mathbf{f}_{l,j}(x_j)$ for some univariate maps $\mathbf{f}_j$ and $\mathbf{f}_{l,j}\}$.
- $\mathcal{M}_{add}$ is a sum space such that $\mathcal{M}_{add} = \mathcal{M}_1 + \cdots \mathcal{M}_d$.

$$\mathcal{M}_j \subset \mathcal{M}_{add} \subset \mathcal{M}$$

Norm on $\mathcal{M}_{add}$

For $\mathbf{f}^{F,\mathrm{tp}} = (\mathbf{f}, \mathbf{f}_{1,1}, \ldots, \mathbf{f}_{r,1}, \ldots, \mathbf{f}_{1,d}, \ldots, \mathbf{f}_{r,d})^\top \in \mathcal{M}_{add}$,

$$\|\mathbf{f}^{F,\mathrm{tp}}\|_{2,*}^2 = \int_0^1 \| \bigoplus_{j=1}^{d} \mathbf{f}_j(x_j)\|^2 p(\mathbf{x}) d\mathbf{x} + \sum_{j=1}^{d}\sum_{l=1}^{r} \|\mathbf{f}_{l,j}(x_j)\|^2 p_j(x_j) dx_j$$

Frechet derivatives of Hilbertian maps

For Hilbertian map $\mathbf{f} : [0,1] \to \mathbb{H}$, $D\mathbf{f}(u) : \mathbb{R} \to \mathbb{H}$ is a bounded linear map from $\mathbb{R}$ to $\mathbb{H}$ such that $D\mathbf{f}(u)(v) = v \odot D\mathbf{f}(u)(1)$, where $D\mathbf{f}(u)(1) \in \mathbb{H}$ is defined by

$$\lim_{|\epsilon|\to 0} \frac{1}{|\epsilon|} \|\mathbf{f}(u+\epsilon) \ominus \mathbf{f}(u) \ominus (\epsilon \odot D\mathbf{f}(u)(1))\| = 0.$$

For $l \geq 2$, the lth-order Fréchet derivative of $\mathbf{f}$ at $u \in [0,1]$, which we denote by $D^l\mathbf{f}(u)$, is a bounded linear map from $\mathbb{R}^l$ to $\mathbb{H}$ such that $D^l\mathbf{f}(u)(v_1, \ldots, v_l) = v_l \odot D^l\mathbf{f}(u)(v_1, \ldots, v_{l-1}, 1)$, where $D^l\mathbf{f}(u)(v_1, \ldots, v_{l-1}, 1) \in \mathbb{H}$ is defined by

$$\lim_{|\epsilon|\to 0} \frac{1}{|\epsilon|} \|D^{l-1}\mathbf{f}(u+\epsilon)(v_1, \ldots, v_{l-1}) \ominus D^{l-1}\mathbf{f}(u)(v_1, \ldots, v_{l-1})$$

$$\ominus(\epsilon \odot D^l\mathbf{f}(u)(v_1, \ldots, v_{l-1}, 1))\| = 0.$$

It holds that $D^l\mathbf{f}(u)(v_1, \ldots, v_l) = v_1 \cdots v_l \odot D^l\mathbf{f}(u)(1, \ldots, 1)$.

Frechet derivatives of Hilbertian maps

If $\mathbf{f} : [0,1] \to \mathbb{H}$ is $(r+1)$-times continuously Fréchet differentiable at $u \in [0,1]$, then the following version of Taylor's expansion holds:

$$\mathbf{f}(v) = \mathbf{f}(u) \oplus \bigoplus_{l=1}^{r} \frac{(v-u)^l}{l!} \odot D^l \mathbf{f}(u)(1, \ldots, 1) \oplus \mathbf{R}(v,u),$$

where $\mathbf{R}(v,u) = ((v-u)^{r+1}/r!) \int_0^1 (1-t)^r \odot D^{r+1}\mathbf{f}(u + t(v-u))(1, \ldots, 1)dt$. The integration in the remainder term is a Bochner integral.

Locally polynomial Hilbertian regression

Using local approximation

$$E(\mathbf{Y}|\mathbf{X}=\mathbf{u}) = \mathbf{m}(\mathbf{u}) \approx \mathbf{m}(\mathbf{x}) \oplus \bigoplus_{j=1}^{d} \bigoplus_{l=1}^{r} \frac{(u_j - x_j)^l}{l!} \odot D^l \mathbf{m}_j(x_j)(1, ..., 1)$$

$$= \mathbf{m}(\mathbf{x}) \oplus \bigoplus_{j=1}^{d} \bigoplus_{l=1}^{r} \left(\frac{u_j - x_j}{h_j} \right)^l \odot \mathbf{m}_{l,j}(x_j)$$

with $\mathbf{m}_{l,j}(\cdot) = \frac{h_j^l}{l!} \odot D^l \mathbf{m}_j(\cdot)(1, ..., 1)$. Minimize

$$\mathcal{L}(\mathbf{f}^{F,\mathrm{tp}}) = \int_{[0,1]^d} n^{-1} \sum_{i=1}^{n} \left\| \mathbf{Y}_i \ominus \bar{\mathbf{Y}} \ominus \mathbf{f}(\mathbf{x}) \ominus \bigoplus_{j=1}^{d} \bigoplus_{l=1}^{r} \left(\frac{X_{ij} - x_j}{h_j} \right)^l \odot \mathbf{f}_{l,j}(x_j) \right\|^2$$

$$\times \prod_{j=1}^{d} K_{h_j}(x_j, X_{ij}) \, d\mathbf{x}$$

with $\mathbf{f}^{F,tp} \in \mathcal{M}_{add}$.

Locally polynomial Hilbertian regression

Let $\mathbf{f}_j^{\mathrm{tp}} = (\mathbf{f}_j, \mathbf{f}_{1,j}, \ldots, \mathbf{f}_{r,j})^\top$, $1 \le j \le d$ be a sub-tuple of $\mathbf{f}^{F,tp}$ and $\mathbf{vec}(u)^T = (1, u, ..., u^r)$.

$$\mathcal{L}(\mathbf{f}^{F,\mathrm{tp}}) = \int_{[0,1]^d} \left\| n^{-1} \sum_{i=1}^n \mathbf{Y}_i \ominus \bar{\mathbf{Y}} \ominus \bigoplus_{j=1}^d \mathbf{vec}\left(\frac{X_{ij} - x_j}{h_j}\right)^\top \odot \mathbf{f}_j^{\mathrm{tp}}(x_j) \right\|^2 \prod_{j=1}^d K_{h_j}(x_j, X_{ij})\, d\mathbf{x}.$$

Then

$$\begin{aligned} D\mathcal{L}(\mathbf{f}^{F,\mathrm{tp}})(\delta^{F,tp}) &:= \lim_{\epsilon \to 0} \epsilon^{-1} \left[\mathcal{L}(\mathbf{f}^{F,\mathrm{tp}} \oplus \epsilon \odot \delta^{F,tp}) - \mathcal{L}(\mathbf{f}^{F,\mathrm{tp}})\right] \\ &= -2 \sum_{j=1}^d \int_0^1 \sum_{l=0}^r \langle \delta_{l,j}(x_j), \mathbf{F}_{l,j}(x_j, \mathbf{f}^{F,\mathrm{tp}}) \rangle \, dx_j. \end{aligned}$$

where

$$\begin{aligned} \mathbf{F}_{l,j}(x_j, \mathbf{f}^{F,\mathrm{tp}}) = \int_{[0,1]^{d-1}} n^{-1} \sum_{i=1}^n \left(\frac{X_{ij} - x_j}{h_j}\right)^l \prod_{k=1}^d K_{h_k}(x_k, X_{ik}) \\ \odot \left[\mathbf{Y}_i \ominus \bar{\mathbf{Y}} \ominus \bigoplus_{k=1}^d \mathbf{vec}\left(\frac{X_{ik} - x_k}{h_k}\right)^\top \odot \mathbf{f}_k^{\mathrm{tp}}(x_k)\right] d\mathbf{x}_{-j}. \end{aligned}$$

Locally polynomial Hilbertian regression

Let $\hat{\mathbf{m}}^{F,\mathrm{tp}} \equiv (\hat{\mathbf{m}}_1 \oplus \cdots \oplus \hat{\mathbf{m}}_d, \hat{\mathbf{m}}_{1,1}, \ldots, \hat{\mathbf{m}}_{r,1}, \ldots, \hat{\mathbf{m}}_{1,d}, \ldots, \hat{\mathbf{m}}_{r,d})$ minimizes $\mathcal{L}(\mathbf{f}^{F,\mathrm{tp}})$.
Then $D\mathcal{L}(\mathbf{f}^{F,\mathrm{tp}})(\delta^{F,tp}) = 0$ for all tuples $\delta^{F,tp}$, which means $\mathbf{F}_{l,j}(x_j, \hat{\mathbf{m}}^{F,tp}) = \mathbf{0}$ for all $0 \le l \le r$ and $1 \le j \le d$.
Define $(r+1) \times (r+1)$ real matrices

$$\hat{\mathbf{M}}_{j,j}(x_j) = n^{-1} \sum_{i=1}^{n} \textbf{vec}\left(\frac{X_{ij} - x_j}{h_j}\right) \textbf{vec}\left(\frac{X_{ij} - x_j}{h_j}\right)^{\top} K_{h_j}(x_j, X_{ij}),$$

$$\hat{\mathbf{M}}_{j,k}(x_j) = n^{-1} \sum_{i=1}^{n} \textbf{vec}\left(\frac{X_{ij} - x_j}{h_j}\right) \textbf{vec}\left(\frac{X_{ik} - x_k}{h_k}\right)^{\top} K_{h_j}(x_j, X_{ij}) K_{h_k}(x_k, X_{ik}).$$

Locally polynomial Hilbertian regression

$\mathbf{F}_{l,j}(x_j, \hat{\mathbf{m}}^{F,tp}) = \mathbf{0}$ for all $0 \le l \le r$ and $1 \le j \le d$ where

$$\mathbf{F}_{l,j}(x_j, \hat{\mathbf{m}}^{F,tp}) = \int_{[0,1]^{d-1}} n^{-1} \sum_{i=1}^{n} \left(\frac{X_{ij} - x_j}{h_j} \right)^l \prod_{k=1}^{d} K_{h_k}(x_k, X_{ik})$$

$$\odot \left[\mathbf{Y}_i \ominus \bar{\mathbf{Y}} \ominus \bigoplus_{k=1}^{d} \textbf{vec} \left(\frac{X_{ik} - x_k}{h_k} \right)^\top \odot \hat{\mathbf{m}}_k^{\mathrm{tp}}(x_k) \right] d\mathbf{x}_{-j}$$

can be written as

$$\hat{\mathbf{m}}_j^{\mathrm{tp}}(x_j) = \tilde{\mathbf{m}}_j^{\mathrm{tp}}(x_j) \odot u_0 \odot \bar{Y} \ominus \bigoplus_{k \neq j} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \cdot \hat{\mathbf{M}}_{jk}(x_j, x_k) \odot \hat{\mathbf{m}}_k^{\mathrm{tp}}(x_k)\, dx_k, \quad 1 \le j \le d.$$

where

$$\tilde{\mathbf{m}}_j^{\mathrm{tp}}(x_j) = \hat{\mathbf{M}}_{jj}(x_j)^{-1} \cdot n^{-1} \odot \left[\sum_{i=1}^{n} \textbf{vec} \left(\frac{X_{ij} - x_j}{h_j} \right) \cdot K_{hj}(x_j, X_{ij}) \odot Y_i \right]$$

Invertibility of $\hat{\mathbf{M}}_{jj}(x_j)$

For $\mathbf{v} = (v_0, v_1, \dots, v_r)^T \in \mathbb{R}^{r+1}$,

$$\mathbf{v}^T \hat{\mathbf{M}}_{jj}(x_j)\mathbf{v} = n^{-1} \sum_{i=1}^{n} \left[v_0 + v_1 \left(\frac{X_{ij} - x_j}{h_j} \right) + \cdots + v_r \left(\frac{X_{ij} - x_j}{h_j} \right)^r \right]^2 K_{h_j}(x_j, X_{ij}).$$

If there exist $(r+1)$ different values of X_{ij} among $1 \le i \le n$ such that $K_{h_j}(x_j, X_{ij}) > 0$, then $\hat{\mathbf{M}}_{jj}(x_j)$ is invertible.

Condition (S). It holds that $c < 1$ and $\inf_{u \in [-c,c]} K(u) > 0$, where

$$c = \max_{1 \le j \le d} h_j^{-1} \max_{1 \le i \le n} \Big\{ X_{(r+1),j} \left(1 - X_{(n-r),j} \right), \frac{1}{2} \left(X_{(r+2),j} - X_{(1),j} \right), \dots,$$

$$\dots, \frac{1}{2} \left(X_{(n),j} - X_{(n-r-1),j} \right) \Big\}$$

where $X_{(1),j} < \cdots < X_{(n),j}$ are the ordered X_{ij} for $1 \le i \le n$.

Condition (S) makes $\hat{\mathbf{M}}_{jj}(x_j)$ invertible for each $x_j \in [0,1]$, $1 \le j \le d$.

Invertibility of $\hat{\mathrm{M}}_{jj}(x_j)$

If $h_j \to 0$ and $nh_j \to \infty$ as $n \to \infty$ and $0 < \inf_{x_j \in [0,1]} p_j(x_j) < \sup_{x_j \in [0,1]} p_j(x_j) < \infty$, then $\mathbb{P}(\hat{\mathbf{M}}_{jj}(x_j)\text{is invertible}) \to 1$.

$$\mathbb{P}\left(\text{There are at most } r \text{ distinct } X_{ij} \in [x_j - h_j, x_j + h_j]\right)$$

$$= \sum_{l=0}^{r} \binom{n}{j} q_{x_j}^{l} (1 - q_{x_j})^{n-l}$$

$$= (1 - q_{x_j})^{n-r} \left[\binom{n}{0} (1 - q_{x)_j} + \binom{n}{1} q_{x_j} (1 - q_{x_j})^{r-1} + \cdots + \binom{n}{r} q_{x_j}^{r} \right]$$

$$\leq \exp\left(-nh_j \cdot p_{\min,\epsilon}(x)\right) \cdot (r+1) \cdot \left(2nh_j \cdot p_{\max,\epsilon}(x)\right)^r for\ all\ large\ n$$

$$\to 0$$

using $\binom{n}{i} \leq n^i, (1-x)^n \leq exp(-nx)$ for $x \geq 0$

Constraints for estimating component maps

Though $\hat{\mathbf{m}}^{F,tp}$ is unique, $\hat{\mathbf{m}}_l^* = \hat{\mathbf{m}}_l \ominus \mathbf{c}$ and $\hat{\mathbf{m}}_{l'}^* = \hat{\mathbf{m}}_{l'} \oplus \mathbf{c}$ for some $1 \leq l \neq l' \leq d$ where $\mathbf{0} \neq \mathbf{c} \in \mathbb{H}$ can make same $\hat{\mathbf{m}}^{F,tp}$.

Recall SBF equations

$$\hat{\mathbf{m}}_j^{\mathrm{tp}}(x_j) = \tilde{\mathbf{m}}_j^{\mathrm{tp}}(x_j) \odot u_0 \odot \bar{Y} \ominus \bigoplus_{k \neq j} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \cdot \hat{\mathbf{M}}_{jk}(x_j, x_k) \odot \hat{\mathbf{m}}_k^{\mathrm{tp}}(x_k)\, dx_k, \quad 1 \leq j \leq d.$$

After computation, we can derive

$$\bigoplus_{j=1}^{d} \int_0^1 \left[\hat{p}_j(x_j) \odot \hat{\mathbf{m}}_j(x_j) \oplus \bigoplus_{l=1}^{r} \hat{p}_{l,j}(x_j) \hat{\mathbf{m}}_{l,j}(x_j) \right] dx_j = \mathbf{0}$$

Constraint $\int_0^1 \hat{p}_j(x_j) \odot \hat{\mathbf{m}}_j(x_j) = \mathbf{0}$ is not inconsistent with SBF equations. And new constraints

$$\int_0^1 \left[\hat{p}_j(x_j) \odot \hat{\mathbf{m}}_j(x_j) \oplus \bigoplus_{l=1}^{r} \hat{p}_{l,j}(x_j) \hat{\mathbf{m}}_{l,j}(x_j) \right] dx_j = \mathbf{0}, \quad 1 \leq j \leq d.$$

Constraints for estimating component maps

where $\hat{p}_j(x_j) = n^{-1} \sum K_{h_j}(x_j, X_{ij})$ and

$$\hat{p}_{l,j}(x_j) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{X_{ij} - x_j}{h_j} \right)^l K_{h_j}(x_j, X_{ij}), \quad 1 \leq l \leq r.$$

Locally polynomial Hilbertian SBF algorithm

Recall SBF equations

$$\hat{\mathbf{m}}_j^{[\ell],\mathrm{tp}}(x_j) = \tilde{\mathbf{m}}_j^{\mathrm{tp}}(x_j) \ominus (u_0 \odot \bar{Y}) \ominus \bigoplus_{k \le j-1} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \cdot \hat{\mathbf{M}}_{jk}(x_j, x_k) \odot \hat{\mathbf{m}}_k^{[\ell],\mathrm{tp}}(x_k) dx_k$$

$$\ominus \bigoplus_{k \ge j+1} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \cdot \hat{\mathbf{M}}_{jk}(x_j, x_k) \odot \hat{\mathbf{m}}_k^{[\ell-1],\mathrm{tp}}(x_k) dx_k \quad 1 \le j \le d.$$

until

$$\|\hat{\mathbf{m}}^{[\ell],F,\mathrm{tp}} \ominus \hat{\mathbf{m}}^{[\ell-1],F,\mathrm{tp}}\|_{2,*}^2 < \epsilon$$

Projection operator

Defnie $(r+1)\times(rd+1)$ matrices by

$$\mathfrak{U}_j = \begin{pmatrix} 1 & \mathbf{0}_{1\times r(j-1)} & \mathbf{0}_{1\times r} & \mathbf{0}_{1\times r(d-j)} \\ \mathbf{0}_{r\times 1} & \mathbf{0}_{r\times r(j-1)} & \mathbf{I}_r & \mathbf{0}_{r\times r(d-j)} \end{pmatrix},$$

where $\mathbf{0}_{k\times k'}$ is the $(k\times k')$ zero matrix and $\mathbf{I}_k$ is the $(k\times k)$ identity matrix. The real matrices $\mathfrak{U}_j$ map $\mathcal{M}_j$ to the spaces where the sub-tuples $\mathbf{f}_j^{F,tp}$ reside. Indeed,

$$\mathfrak{U}_j \odot \mathbf{f}_j^{F,tp} = \mathbf{f}_j^{F,tp}, \quad \mathbf{f}_j^{F,tp} = \mathfrak{U}_j \odot \mathbf{f}^{F,tp}.$$

Define the operators $\hat{\pi}_j : \mathcal{M} \to \mathcal{M}_j$ by

$$\hat{\pi}_j(\mathbf{f}^{F,tp}) = \mathfrak{U}_j^\mathsf{T} \odot \hat{\mathbf{g}}_j^{*,tp} \in \mathcal{M}_j,$$

$$\hat{\mathbf{g}}_j^{*,tp}(x_i) = \hat{\mathbf{M}}_{jj}(x_i)^{-1} \odot \int_{[0,1]^{d-1}} \mathfrak{U}_j \cdot \hat{\mathfrak{M}}^F(\mathbf{x}) \odot \mathbf{f}^{F,tp}(\mathbf{x})\, d\mathbf{x}_{-j}.$$

where

$$\hat{\mathfrak{M}}^F(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} vec((\mathbf{X_i} - \mathbf{x}/\mathbf{h}) \cdot vec((\mathbf{X_i} - \mathbf{x}/\mathbf{h})^T \cdot \prod_{l=1}^{d} K_{h_l}(x_l, X_{il})$$

Projection operator

Replacing $\hat{\mathbf{M}}_{jj}(x_j)$ and $\hat{\mathfrak{M}}^F(\mathbf{x})$ by $\mathbf{N} \cdot p_j(x_j)$ and $\mathfrak{N}^F p(\mathbf{x})$, respectively. Define π_j by

$$\pi_j(\mathbf{f}^{F,tp}) = \mathfrak{U}_j^\top \odot \mathbf{g}_j^{*,tp} \in \mathcal{M}_j,$$

$$\mathbf{g}_j^{*,tp}(x_j) = \mathbf{N}^{-1} \cdot \mathfrak{U}_j \cdot \mathfrak{N}^F \odot \int_{[0,1]^{d-1}} \frac{p(\mathbf{x})}{p_j(x_j)} \odot \mathbf{f}^{F,tp}(\mathbf{x})\, d\mathbf{x}_{-j}.$$

where

$$\mathfrak{N}^F = \int_{-1}^{1} vec(\mathbf{u}) \cdot vec(\mathbf{u})^T \prod_{j=1}^{d} K(u_j) d\mathbf{u}$$

Projection operator

Define $\|\cdot\|_2 : \mathcal{M} \to [0,\infty)$ by $\|\mathbf{f}^{F,tp}\|_2^2 = \langle \mathbf{f}^{F,tp}, \mathbf{f}^{F,tp} \rangle_2$ where

$$\langle \mathbf{f}^{F,tp}, \mathbf{g}^{F,tp} \rangle_2 = \int_{[0,1]^d} \langle \mathbf{f}^{F,tp}(\mathbf{x}), \mathfrak{N}^F \odot \mathbf{g}^{F,tp}(\mathbf{x}) \rangle_{\mathbb{H}^{rd+1}} p(\mathbf{x}) d\mathbf{x}, \quad \mathbf{f}^{F,tp}, \mathbf{g}^{F,tp} \in \mathcal{M}. \tag{A.2}$$

First we observe that $\mathfrak{N}^F$ is positive definite. For $\mathbf{c} \in \mathbb{R}^{d+1}$,

$$\mathbf{c}^\top \mathfrak{N}^F \mathbf{c} = \int_{[0,1]^d} \left(c_0 + \sum_{j=1}^{d} \sum_{l=1}^{r} c_{l,j} u_j^l \right)^2 \prod_{\ell=1}^{d} K(u_\ell)\, d\mathbf{u}$$

and that K is supported on $[-1,1]$.

Projection operator

Now, decompose $\mathfrak{N}^F = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}^\top$. Then,

$$\lambda_{\min} \left\| \mathbf{f}^{\mathbf{F},tp} \right\|^2_{\mathbb{H}^{rd+1}} \le \left\langle \mathbf{f}^{\mathbf{F},tp}, \mathfrak{N}^F \odot \mathbf{f}^{\mathbf{F},tp} \right\rangle_{\mathbb{H}^{rd+1}} = \left\| \boldsymbol{\Lambda}^{1/2}\mathbf{P}^\top \odot \mathbf{f}^{\mathbf{F},tp} \right\|^2_{\mathbb{H}^{rd+1}} \le \lambda_{\max} \left\| \mathbf{f}^{\mathbf{F},tp} \right\|^2_{\mathbb{H}^{rd+1}},$$

Furthermore,

$$\begin{aligned} \lambda_{\max}^{-1} \|\mathbf{f}^{F,tp}\|_2^2 &\le \int_{[0,1]^d} \|\mathbf{f}^0(\mathbf{x})\|^2 p(\mathbf{x}) d\mathbf{x} + \sum_{j=1}^{d} \sum_{l=1}^{r} \int_{[0,1]^d} \|\mathbf{f}^j(\mathbf{x})\|^2 p(\mathbf{x}) d\mathbf{x} = \|\mathbf{f}^{F,tp}\|_{2,*}^2 \\ &\le \lambda_{\min}^{-1} \|\mathbf{f}^{F,tp}\|_2^2 \end{aligned}$$

Projection operator

We also get

$$\left\langle \mathbf{f}^{F,tp}, \mathfrak{N}^F \circ \mathbf{g}^{F,tp} \right\rangle_{\mathbb{H}^{dr+1}} \leq \left|\left\langle \mathbf{f}^{\mathbf{F},tp}, \mathfrak{N}^F \odot \mathbf{f}^{\mathbf{F},tp} \right\rangle_{\mathbb{H}^{dr+1}}\right|^{1/2} \cdot \left|\left\langle \mathbf{g}^{\mathbf{F},tp}, \mathfrak{N}^F \odot \mathbf{g}^{\mathbf{F},tp} \right\rangle_{\mathbb{H}^{dr+1}}\right|^{1/2}.$$

This with an application of the Hölder inequality gives

$$\begin{aligned}\left\|\left\langle \mathbf{f}^{\mathbf{F},tp}, \mathbf{g}^{\mathbf{F},tp} \right\rangle_2\right\| &\leq \int_{[0,1]^d} \left|\left\langle \mathbf{f}^{\mathbf{F},tp}(\mathbf{x}), \mathfrak{N}^F \odot \mathbf{f}^{\mathbf{F},tp}(\mathbf{x}) \right\rangle_{\mathbb{H}^{dr+1}}\right|^{1/2} \cdot \left|\left\langle \mathbf{g}^{\mathbf{F},tp}(\mathbf{x}), \mathfrak{N}^F \odot \mathbf{g}^{\mathbf{F},tp}(\mathbf{x}) \right\rangle_{\mathbb{H}^{dr+1}}\right|^{1/} \\ &\leq \left\|\mathbf{f}^{\mathbf{F},tp}\right\|_2 \cdot \left\|\mathbf{g}^{\mathbf{F},tp}\right\|_2.\end{aligned}$$

Projection operator

Check that $\langle \mathbf{f}^{F,tp} \ominus \pi_j(\mathbf{f}^{F,tp}), \mathbf{g}_j^{F,tp} \rangle_2 = 0$, for all $\mathbf{g}_j^{F,tp} \in \mathcal{M}_j$.

$$\begin{aligned}
&\left\langle \mathbf{f}^{F,\mathrm{tp}}(\mathbf{x}) \ominus (\mathfrak{U}_j^{\mathrm{T}} \odot \mathbf{g}_j^{*,\mathrm{tp}}(x_j)), \mathfrak{N}^{\mathrm{F}} \odot \mathbf{g}_j^{F,\mathrm{tp}}(x_j) \right\rangle_{\mathbb{H}^{rd+1}} \\
&= \left\langle \mathfrak{N}^{F} \odot (\mathbf{f}^{F,\mathrm{tp}}(\mathbf{x}) \ominus (\mathfrak{U}_j^{\mathrm{T}} \odot \mathbf{g}_j^{*,\mathrm{tp}}(x_j))), \mathbf{g}_j^{F,\mathrm{tp}}(x_j) \right\rangle_{\mathbb{H}^{rd+1}} \\
&= \left\langle \mathfrak{U}_j \cdot \mathfrak{N}^{\mathrm{F}} \odot (\mathbf{f}^{F,\mathrm{tp}}(\mathbf{x}) \ominus (\mathfrak{U}_j^{\mathrm{T}} \odot \mathbf{g}_j^{*,\mathrm{tp}}(x_j))), \mathbf{g}_j^{F,\mathrm{tp}}(x_j) \right\rangle_{\mathbb{H}^{d+1}}
\end{aligned}$$

We have used

$$\langle f, A \odot g \rangle_{\mathbb{H}^{rd+1}} = \sum_{l=1}^{rd+1} \langle f_l, \bigoplus_{k=1}^{rd+1} (A_{l,k} \odot g_k) \rangle_{\mathbb{H}} = \sum_{k=1}^{rd+1} \langle \bigoplus_{l=1}^{rd+1} (A_{l,k} \odot f_l)\, , g_k \rangle_{\mathbb{H}} = \langle A^T \odot f, g \rangle_{\mathbb{H}^{rd+1}}$$

The second equality follows due to the structure of $\mathbf{g}_j^{F,\mathrm{tp}}$.

Projection operator

Using $\mathfrak{U}_j \mathfrak{N}^F \mathfrak{U}_j^T = \mathbf{N}$,

$$\left\langle \mathbf{f}^{F,\text{tp}} \ominus \pi_j(\mathbf{f}^{F,\text{tp}}), \mathbf{g}_j^{F,\text{tp}} \right\rangle_2$$

$$= \int_{[0,1]^d} \left\langle \mathfrak{U}_j \cdot \mathfrak{N}^{\text{F}} \odot \mathbf{f}^{F,\text{tp}}(\mathbf{x}) \ominus \mathbf{N} \odot \mathbf{g}_j^{*,\text{tp}}(x_j), \mathbf{g}_j^{\text{tp}}(x_j) \right\rangle_{\mathbb{H}^{d+1}} \cdot p(\mathbf{x}) \mathbf{dx}$$

$$= \int_0^1 \left\langle \mathfrak{U}_j \cdot \mathfrak{N}^{\text{F}} \odot \int_{[0,1]^{d-1}} \left(\mathbf{f}^{F,\text{tp}}(\mathbf{x}) \ominus \int_{[0,1]^{d-1}} \frac{p(\mathbf{x})}{p_j(x_j)} \odot \mathbf{f}^{F,\text{tp}}(\mathbf{x}) \mathbf{dx}_{-j} \right) \right.$$

$$\left. \odot p(\mathbf{x}) \mathbf{dx}_{-j}, \mathbf{g}_j^{\text{tp}}(x_j) \right\rangle_{\mathbb{H}^{d+1}} dx_j = 0$$

since $\int_{[0,1]^{d-1}} \left(\mathbf{f}^{F,\text{tp}}(\mathbf{x}) \ominus \int_{[0,1]^{d-1}} \frac{p(\mathbf{x})}{p_j(x_j)} \odot \mathbf{f}^{F,\text{tp}}(\mathbf{x}) \mathbf{dx}_{-j} \right) p(\mathbf{x}) d\mathbf{x}_{\text{-}j} = \mathbf{0}_{rd+1}$

Projection operator

If we restrict the domain of $\hat{\pi}_j$to $\mathcal{M}_k, j \neq k$, $\mathbf{f}_k^{F,\mathrm{tp}} = (\mathbf{f}_k, \mathbf{0}_{r(k-1)}^{\mathrm{T}}, \mathbf{f}_{1,k}, \ldots, \mathbf{f}_{r,k}, \mathbf{0}_{r(d-k)}^{\mathrm{T}}) \in \mathcal{M}_k$,

$$(\hat{\pi}_j|\mathcal{M}_k)(\mathbf{f}_k^{F,\mathrm{tp}}) = \mathfrak{U}_j^{\mathrm{T}} \odot \mathbf{g}_j^{*,\mathrm{tp}} \in \mathcal{M}_j,$$

$$\hat{\mathbf{g}}_j^{*,tp}(x_i) = \hat{\mathbf{M}}_{jj}(x_i)^{-1} \odot \int_{[0,1]^{d-1}} \mathfrak{U}_j \cdot \hat{\mathfrak{M}}^F(\mathbf{x}) \odot \mathbf{f}^{F,tp}(\mathbf{x})\, d\mathbf{x}_{-j}.$$

$$= \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \hat{\mathbf{M}}_{jk}(x_j, x_k) \odot \mathbf{f}_k^{F,\mathrm{tp}}(x_k) dx_k.$$

Projection operator

Similarly, we get $\pi_j(\mathbf{f}^{F,\mathrm{tp}})$ and $(\pi_j|_{\mathcal{M}_k})(\mathbf{f}_k^{F,\mathrm{tp}})$ for $\mathbf{f}^{F,\mathrm{tp}} \in \mathcal{M}_{\mathrm{add}}$ and $\mathbf{f}_k^{F,\mathrm{tp}} \in \mathcal{M}_k$. Namely, $\pi_j(\mathbf{f}^{F,\mathrm{tp}})$ is given as at (A.6) with $\mathbf{g}_j^{*,\mathrm{tp}}$ being replaced by

$$\mathbf{g}_j^{*,\mathrm{tp}}(x_j) = \mathbf{f}_j^{\mathrm{tp}}(x_j) \oplus \mathbf{N}^{-1} \cdot \boldsymbol{\mu} \cdot \boldsymbol{\mu}^{\mathrm{T}} \odot \bigoplus_{k \neq j} \int_0^1 \frac{p_{jk}(x_j, x_k)}{p_j(x_j)} \odot \mathbf{f}_k^{\mathrm{tp}}(x_k) dx_k,$$

where $\boldsymbol{\mu} = \int_{-1}^1 \mathrm{vec}(u) K(u) du$. Also, we obtain

$$(\pi_j|_{\mathcal{M}_k})(\mathbf{f}_k^{F,\mathrm{tp}}) = \mathfrak{U}_j^{\mathrm{T}} \odot \mathbf{g}_j^{*,\mathrm{tp}} \in \mathcal{M}_j,$$

$$\mathbf{g}_j^{*,\mathrm{tp}}(x_j) = \mathbf{N}^{-1} \cdot \boldsymbol{\mu} \cdot \boldsymbol{\mu}^{\mathrm{T}} \odot \int_0^1 \frac{p_{jk}(x_j, x_k)}{p_j(x_j)} \odot \mathbf{f}_k^{\mathrm{tp}}(x_k) dx_k.$$

Existence of estimators and convergence of algorithm

(A1) The joint density p is bounded away from zero and infinity on $[0,1]^d$, and p_{jk} are continuous on $[0,1]^2$ for $1 \leq j \neq k \leq d$.
(A2) The baseline kernel K is Lipschitz continuous.

Theorem 1. Assume that the conditions (A1)–(A2) hold and $\mathbb{E}(\|\mathbf{Y}\|^\alpha) < \infty$ for some $\alpha > 2$. Also, assume that $h_j = o(1)$ as $n \to \infty$, and that there exist $0 < c_j < (\alpha - 2)/\alpha$ with $c_j + c_k < 1$ for all $1 \leq j \neq k \leq d$ such that $n^{c_j} h_j$ are bounded away from zero for all $1 \leq j \leq d$. Then, with probability tending to one, there exists a unique tuple $\hat{\mathbf{m}}^{F,\mathsf{tp}}$ that satisfies the SBF equations. Also, provided that $\|\hat{\mathbf{m}}^{[0],F,\mathsf{tp}}\|_{2,*} \leq C'$ with probability tending to one for some $0 < C' < \infty$, there exist some constants $0 < C < \infty$ and $0 < \gamma < 1$, such that $\|\hat{\mathbf{m}}^{[\ell],F,\mathsf{tp}} \ominus \hat{\mathbf{m}}^{F,\mathsf{tp}}\|_{2,*} \leq C \cdot \gamma^\ell$ for all $\ell \geq 1$ with probability tending to one.

Existence of estimators and convergence of algorithm

Recall SBF equations

$$\hat{\mathbf{m}}_j^{\mathrm{tp}}(x_j) = \tilde{\mathbf{m}}_j^{\mathrm{tp}}(x_j) \odot u_0 \odot \bar{Y} \ominus \bigoplus_{k \neq j} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \cdot \hat{\mathbf{M}}_{jk}(x_j, x_k) \odot \hat{\mathbf{m}}_k^{\mathrm{tp}}(x_k)\, dx_k, \quad 1 \leq j \leq d.$$

and

$$\hat{\pi}_j(\mathbf{f}_k^{F,tp}) = \mathfrak{U}_j^T \odot \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \hat{\mathbf{M}}_{jk}(x_j, x_k) \odot \mathbf{f}_k^{F,\mathrm{tp}}(x_k) dx_k, \quad 1 \leq j \neq k \leq d.$$

We get that

$$\hat{\mathbf{m}}^{F,\mathrm{tp}} = \tilde{\mathbf{m}}_j^{F,\mathrm{tp}} \oplus (I - \hat{\pi}_j)\hat{\mathbf{m}}^{F,\mathrm{tp}},$$

$$\bigoplus_{k=1}^{j} \hat{\mathbf{m}}_k^{[\ell],F,\mathrm{tp}} \oplus \bigoplus_{k=j+1}^{d} \hat{\mathbf{m}}_k^{[\ell-1],F,\mathrm{tp}} = \tilde{\mathbf{m}}_j^{F,\mathrm{tp}} \oplus (I - \hat{\pi}_j) \left(\bigoplus_{k=1}^{j-1} \hat{\mathbf{m}}_k^{[\ell],F,\mathrm{tp}} \oplus \bigoplus_{k=j}^{d} \hat{\mathbf{m}}_k^{[\ell-1],F,\mathrm{tp}} \right)$$

for $1 \leq j \leq d$.

Existence of estimators and convergence of algorithm

Applying the above systems of equations backward from $j = d$ to $j = 1$ successively, we get

$$\hat{\mathbf{m}}^{F,\mathrm{tp}} = \hat{T}\hat{\mathbf{m}}^{F,\mathrm{tp}} \oplus \tilde{\mathbf{m}}_{\oplus}^{F,\mathrm{tp}},$$

$$\hat{\mathbf{m}}^{[\ell],F,\mathrm{tp}} = \hat{T}\hat{\mathbf{m}}^{[\ell-1],F,\mathrm{tp}} \oplus \tilde{\mathbf{m}}_{\oplus}^{F,\mathrm{tp}},$$

where
$\mathbf{m}_{\oplus}^{F,\mathrm{tp}} = \mathbf{m}_{d}^{F,\mathrm{tp}} \oplus (I - \hat{\pi}_d)\mathbf{m}_{d-1}^{F,\mathrm{tp}} \oplus (I - \hat{\pi}_d)(I - \hat{\pi}_{d-1})\mathbf{m}_{d-2}^{F,\mathrm{tp}} \oplus \cdots \oplus (I - \hat{\pi}_d)\cdots(I - \hat{\pi}_2)\mathbf{m}_{1}^{F,\mathrm{tp}} = (I - \hat{T})\mathbf{m}^{\mathbf{F},\mathbf{tp}}$. By the contraction mapping theorem (Chapter 4, Sacks (2017)), SBF equations have a unique solution $\mathbf{m}^{F,\mathrm{tp}}$ if $\|\hat{T}\|_{\mathcal{L}(\mathcal{M}_{\mathsf{add}},\mathcal{M}_{\mathsf{add}})}\| < 1$. We claim

$$P(\|\hat{T}\|_{\mathcal{L}(\mathcal{M}_{\mathsf{add}},\mathcal{M}_{\mathsf{add}})} < \gamma) \to 1$$

$$\|\mathbf{m}^{[\ell],F,\mathrm{tp}} \ominus \hat{\mathbf{m}}^{F,\mathrm{tp}}\|_2 \le \frac{\|\hat{T}\|^{\ell}_{\mathcal{L}(\mathcal{M}_{\mathrm{add}},\mathcal{M}_{\mathrm{add}})}}{1 - \|\hat{T}\|_{\mathcal{L}(\mathcal{M}_{\mathrm{add}},\mathcal{M}_{\mathrm{add}})}} \cdot \left(\|\mathbf{m}^{[0],F,\mathrm{tp}}\|_2 + \|\tilde{\mathbf{m}}_{\oplus}^{F,\mathrm{tp}}\|_2\right).$$

Error rates

(A3) The true component maps m_j for $1 \leq j \leq d$ are (r + 1)-times continuously Fréchet differentiable on $[0, 1]$.
(A4) The bandwidths h_j satisfy $n^{1/(2r+3)} h_j \to \alpha_j$ for some constants $\alpha_j > 0, 1 \leq j \leq d$.

$$N_j(u) = \int_0^1 \text{vec}\left(\frac{v-u}{h_j}\right) \cdot \text{vec}\left(\frac{v-u}{h_j}\right)^\top K_h(u,v)\,dv,$$

$$\gamma_j(u) = \int_0^1 \left(\frac{v-u}{h_j}\right)^{r+1} \cdot \text{vec}\left(\frac{v-u}{h_j}\right) K_h(u,v)\,dv.$$

We note that for u in the interior region $I_j := [2h_j, 1-2h_j]$, the above $N_j(u)$ and $\gamma_j(u)$ reduce to constant $(r+1) \times (r+1)$-matrix and $(r+1)$-vector, respectively, that involve complete moments of K. Indeed, for all $u \in I_j$, it holds that $N_j(u) = N$ and $\gamma_j(u) = \gamma$, where

$$N = \int_{-1}^1 \text{vec}(v) \cdot \text{vec}(v)^\top K(v)\,dv, \quad \gamma = \int_{-1}^1 v^{r+1} \cdot \text{vec}(v) \cdot K(v)\,dv.$$

Error rates

Define

$$\hat{\boldsymbol{\kappa}}_{ij}(x_j) = \hat{\mathbf{M}}_{jj}(x_j)^{-1} \cdot n^{-1}\textbf{vec}(\frac{X_{ij} - x_j}{h_j})K_{h_j}(x_j, X_{ij})$$

$$\tilde{\mathbf{m}}_j^{A,\mathsf{tp}}(x_j) = \bigoplus_{i=1}^{n} \hat{\boldsymbol{\kappa}}_{ij}(x_j) \odot \epsilon_i$$

Then

$$\tilde{\mathbf{m}}_j^{\mathrm{tp}}(x_j) = \hat{\mathbf{M}}_{jj}(x_j)^{-1} \cdot n^{-1} \odot \left[\sum_{i=1}^{n} \textbf{vec}\left(\frac{X_{ij} - x_j}{h_j}\right) \cdot K_{hj}(x_j, X_{ij}) \odot \mathbf{Y}_i\right]$$

$$= \bigoplus_{i=1}^{n} \hat{\boldsymbol{\kappa}}_{ij}(x_j) \odot \mathbf{Y}_i = \bigoplus_{i=1}^{n} \hat{\boldsymbol{\kappa}}_{ij}(x_j) \odot \left(\bigoplus_{k=1}^{d} \mathbf{m}_k(\mathbf{X}_{ik}) \oplus \epsilon_i\right)$$

Error rates

Theorem 2. Assume that the conditions (A1)-(A4) hold, and that $E(\|\epsilon_j\|^\alpha) < \infty$ for some $\alpha > (2r+3)/(r+1)$ and $E(\|\epsilon_j|X_j\|)$ are bounded on $[0,1]$ for all $1 \le j \le d$. Suppose that $\hat{m}_j = (\hat{m}_j, \hat{m}_1, \ldots, \hat{m}_r)^\top$ satisfy the constraints in (3.6). Then, for all $1 \le j \le d$, it holds that

$$\hat{\mathbf{m}}_j^{tp}(x_j) \ominus \mathbf{m}_j^{tp}(x_j) = \tilde{\mathbf{m}}_j^{A,\mathsf{tp}}(x_j) \oplus \frac{h^{r+1}}{(r+1)!}\mathbf{N}_j(x_j)\boldsymbol{\gamma}_j(x_j) \odot D^{r+1}\mathbf{m}_j(x_j)(1,...,1)$$
$$\oplus o_p\left(n^{-(r+1)/(2r+3)}\right)$$

uniformly for $x_j \in [0,1]$.
where we denote $\mathbf{Z}_n = o_p(1)$ for random element $\mathbf{Z}_n : \Omega \to \mathbb{H}^{r+1}$ if $P(\|\mathbf{Z}_n\|_{tp}) > \epsilon) \to 0$ as $n \to \infty$ for any $\epsilon > 0$, define inner product and associated norm for $\mathbb{H}^{r+1}$ as follows.
For $\mathbf{v} = (\mathbf{v}_0, ..., \mathbf{v}_r)^T, \mathbf{w} = (\mathbf{w}_0, ..., \mathbf{w}_r)^T$ in $\mathbb{H}^{r+1}$, $\langle \mathbf{v}, \mathbf{w}\rangle_{tp} = \sum_{j=0}^r \langle \mathbf{v}_j, \mathbf{w}_j\rangle$ and $\|\mathbf{v}\|_{tp} = \sum_{j=0}^r \|\mathbf{v}_j\|^2$.

proof of error rates

Let $\tau_n = n^{-(r+1)/(2r+3)}$
Define

$$\tilde{\mathbf{m}}_j^{B,\mathsf{tp}}(x_j) = \frac{1}{n} \bigoplus_{i=1}^{n} \hat{\boldsymbol{\kappa}}_{ij}(x_j) \odot \left\{ \mathbf{m}_j(X_{ij}) \ominus \textbf{vec} \left(\frac{X_{ij} - x_j}{h_j} \right)^T \odot \mathbf{m}_j^{\mathsf{tp}}(x_j) \right\}, \quad 1 \leq j \leq d.$$

$$\tilde{\mathbf{m}}_{jk}^{C,\mathsf{tp}}(x_j) = \frac{1}{n} \bigoplus_{i=1}^{n} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j) \hat{\boldsymbol{\kappa}}_{ij}(x_j) K_h(x_j, X_{ik})$$

$$\odot \left\{ \mathbf{m}_k(x_k) \ominus \textbf{vec} \left(\frac{X_{ik} - x_k}{h_k} \right)^T \odot \mathbf{m}_k^{\mathsf{tp}}(x_k) \right\} dx_k, \quad 1 \leq j \neq k \leq d$$

proof of error rates

Recall backfitting equations

$$\hat{\mathbf{m}}_j^{\mathrm{tp}}(x_j) = \tilde{\mathbf{m}}_j^{\mathrm{tp}}(x_j) \odot u_0 \odot \bar{Y} \ominus \bigoplus_{k \neq j} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \cdot \hat{\mathbf{M}}_{jk}(x_j, x_k) \odot \hat{\mathbf{m}}_k^{\mathrm{tp}}(x_k)\, dx_k, \quad 1 \leq j \leq d.$$

can be written as

$$\hat{\mathbf{m}}_j^{\mathsf{tp}}(x_j) = \mathbf{m}_j^{\mathsf{tp}}(x_j) \oplus \tilde{\mathbf{m}}_j^{A,\mathsf{tp}}(x_j) \oplus \tilde{\mathbf{m}}_j^{B,\mathsf{tp}}(x_j) \oplus \bigoplus_{k \neq j} \hat{\mathbf{M}}_{jj}(x_j)^{-1} \odot \tilde{\mathbf{m}}_{jk}^{C,\mathsf{tp}}(x_j)$$

$$\ominus \bigoplus_{k \neq j} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \hat{\mathbf{M}}_{jk}(x_j, x_k) \odot \left(\hat{\mathbf{m}}_k^{tp}(x_k) \ominus \mathbf{m}_k^{\mathsf{tp}}(x_k) \right) dx_k, \quad 1 \leq j \leq d.$$

proof of error rates

Claim

$$\sup_{u\in[0,1]}\left|\hat{\mathbf{M}}_{jj}(u) - \mathbf{N}_j(u)\cdot p_j(u)\right| = o_p(1),$$

$$\sup_{(u,v)\in[0,1]^2}\left|\hat{\mathbf{M}}_{jk}(u,v) - \boldsymbol{\mu}_j(u)\cdot\boldsymbol{\mu}_k(v)^\top\cdot p_{jk}(u,v)\right| = o_p(1).$$

We can check $nh_j/(\log n)^3 \to \alpha_j n^{(2r+2)/(2r+3)}/(\log n)^3 \to \infty$ as $n\to\infty$.

proof of error rates

Using Lemma 6.2,

$$\sup_{x_j\in[0,1]}\left|\frac{1}{n}\sum_{i=1}^{n}\left(\frac{X_{ij}-x_j}{h_j}\right)^l K_{h_j}(x_j,X_{ij})-E\left[\left(\frac{X_j-x_j}{h_j}\right)^l K_{h_j}(x_j,X_j)\right]\right| = O_p(\frac{\sqrt{\log n}}{\sqrt{nh_j}})$$

Since p_j is continuous on $[0,1]$,

$$\sup_{x_j\in[0,1]}\left|E\left[\left(\frac{X_j-x_j}{h_j}\right)^l K_{h_j}(x_j,X_j)\right]-\left(\int_0^1\left(\frac{v-x_j}{h_j}\right)^l K_{h_j}(x_j,v)dv\right)p_j(x_j)\right| = o_p(1)$$

We denote $(\boldsymbol{\mu}_j(u))_p=\int_0^1\left(\frac{v-u}{h_j}\right)^p K_{h_j}(u,v)dv, \quad 0 \ \le p\le r$; moments defined with normalized kernel.

proof of error rates

Using multivariate version of Lemma 6.2,

$$\sup_{x_j,x_k\in[0,1]^2}|\frac{1}{n}\sum_{i=1}^{n}\left(\frac{X_{ij}-x_j}{h_j}\right)^l\left(\frac{X_{ik}-x_k}{h_k}\right)^{l'}K_{h_j}(x_j,X_{ij})K_{h_k}(x_k,X_{ik})$$

$$-E\left[\left(\frac{X_j-x_j}{h_j}\right)^l\left(\frac{X_k-x_k}{h_k}\right)^{l'}K_{h_j}(x_j,X_j)K_{h_k}(x_k,X_k)\right]| = O_p(\frac{\sqrt{\log n}}{\sqrt{nh_j}}) = o_p(1)$$

Since $p_{j,k}$ is continuous on $[0,1]^2$,

$$\sup_{x_j,x_k\in[0,1]^2}|E\left[\left(\frac{X_j-x_j}{h_j}\right)^l\left(\frac{X_k-x_k}{h_k}\right)^{l'}K_{h_j}(x_j,X_j)K_{h_k}(x_k,X_k)\right]$$

$$-(\boldsymbol{\mu}_j(x_j))_l(\boldsymbol{\mu}_j(x_k))_{l'}p_{j,k}(x_j,x_k)| = o_p(1)$$

proof of error rates

Then we get

$$\int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \cdot \hat{\mathbf{M}}_{jk}(x_j, x_k) \cdot \mathbf{N}_k(x_k)^{-1} \cdot \boldsymbol{\gamma}_k(x_k) \odot D^{r+1}\mathbf{m}_k(x_k)(1, \ldots, 1)dx_k$$

$$= \int_0^1 \mathbf{N}_k(x_k)^{-1}\boldsymbol{\mu}_j(x_j)\boldsymbol{\mu}_k(x_k)^T\mathbf{N}_k(x_k)^{-1}\boldsymbol{\gamma}_k(x_k)\cdot\frac{p_{j,k}(x_j, x_k)}{p_j(x_j)}\odot D^{r+1}\mathbf{m}_k(x_k)(1, ..., 1)dx_k \oplus o_p(1)$$

$$=$$

$$= \mu_{r+1} \cdot \mathfrak{u}_0 \odot \int_0^1 \frac{p_{jk}(x_j, x_k)}{p_j(x_j)} \odot D^{r+1}\mathbf{m}_k(x_k)(1, \ldots, 1)dx_k \oplus o_p(1).$$

proof of error rates

Using

$$\mathbf{m}_k(X_{ik}) \ominus vec(\frac{X_{ik} - x_k}{h_k})^T \odot \mathbf{m}_k^{tp}(x_k) = \mathbf{m}_k(X_{ik}) \ominus \bigoplus_{l=0}^{r} \frac{(X_{ik} - x_k)^l}{l!} \odot D^l \mathbf{m}_k(x_k)(1, ..., 1)$$

$$= \frac{(X_{ik} - x_k)^{r+1}}{(r+1)!} \odot D^{r+1} \mathbf{m}_k(\xi_{ik})(1, ..., 1)$$

we get

$$\hat{\mathbf{M}}_{jj}(x_j)^{-1} \odot \tilde{\mathbf{m}}_{jk}^{\mathrm{C,tp}}(x_j)$$

$$= \bigoplus_{i=1}^{n} \hat{\kappa}_{ij}(x_j) \odot \int_0^1 \left[\int_0^1 \frac{(v - x_k)^{r+1}}{(r+1)!} \cdot \frac{p_{jk}(X_{ij}, v)}{p_j(X_{ij})} \cdot K_{h_k}(x_k, v) dv \right] \odot D^{r+1} \mathbf{m}_k(x_k)(1, \ldots, 1) dx_k$$

$$\oplus o_p(\tau_n)$$

$$= \frac{\mu_{r+1}}{(r+1)!} \cdot h_k^{r+1} \cdot \mathfrak{u}_0 \odot \int_0^1 \frac{p_{jk}(x_j, x_k)}{p_j(x_j)} \odot D^{r+1} \mathbf{m}_k(x_k)(1, \ldots, 1) dx_k \oplus o_p(\tau_n).$$

proof of error rates

$$(\tilde{\mathbf{m}}_j^{A,tp}(x_j))_l = \hat{\mathbf{M}}_{jj}(x_j)^{-1} \frac{1}{n} \bigoplus_{i=1}^{n} \left(\frac{X_{ij} - x_j}{h_j} \right)^l K_{h_j}(x_j, X_{ij}) \odot \epsilon_i$$

$$E\left[\tilde{\mathbf{m}}_j^{A,tp}(x_j))_l \right] = \mathbf{0} \in \mathbb{H}$$

$$E\left[\|\tilde{\mathbf{m}}_j^{A,tp}(x_j))_l\|^2 \right] = \frac{1}{n} E\left[\left(\frac{X_{ij} - x_j}{h_j} \right)^{2l} K_{h_j}(x_j, X_{ij})^2 \odot \|\epsilon\|^2 \right] \leq \frac{C}{nh_j}$$

for some $C > 0$ with probability tending to 1.
So $(\tilde{\mathbf{m}}_j^{A,tp}(x_j))_l = O_p(\frac{1}{\sqrt{nh_j}}), 0 \leq l \leq r$.

proof of error rates

We can check $E(\epsilon_i|\mathbf{X}_i) = \mathbf{0}$, $h_j \sim n^{-1/(2r+3)}$ with $1/(2r+3) < (\alpha - 2)/\alpha$ which is equivalent to $\alpha > (2r+3)/(r+1)$. And we have $E(\|\epsilon\|^2|X_j = \cdot)$ is bounded on $[0,1]$.
Using Lemma 6.1,

$$\sup_{x_j \in [0,1]} (\tilde{\mathbf{m}}_j^{A,tp}(x_j))_l = O_p(\frac{\sqrt{\log n}}{\sqrt{nh_j}}), \quad 0 \leq l \leq r.$$

Error rates

Recall

$$\hat{\mathbf{m}}_j^{tp}(x_j) \ominus \mathbf{m}_j^{tp}(x_j) = \tilde{\mathbf{m}}_j^{A,\mathsf{tp}}(x_j) \oplus \frac{h^{r+1}}{(r+1)!}\mathbf{N}_j(x_j)\boldsymbol{\gamma}_j(x_j) \odot D^{r+1}\mathbf{m}_j(x_j)(1, ..., 1)$$

$$\oplus o_p\left(n^{-(r+1)/(2r+3)}\right)$$

we get

$$\hat{\mathbf{m}}_j^{\mathrm{tp}}(x_j) \ominus \mathbf{m}_j^{\mathrm{tp}}(x_j) = O_p(n^{-(r+1)/(2r+3)}), \quad x_j \in [0,1];$$

$$\sup_{x_j \in [0,1]} \|\hat{\mathbf{m}}_j^{\mathrm{tp}}(x_j) \ominus \mathbf{m}_j^{\mathrm{tp}}(x_j)\|_{\mathrm{tp}} = O_p(n^{-(r+1)/(2r+3)}\sqrt{\log n}).$$

Error rates

For each component $\mathbf{m}_j$

$$\hat{\mathbf{m}}_j(x_j) \ominus \mathbf{m}_j(x_j) = O_p(n^{-(r+1)/(2r+3)})$$

Using $\frac{l!}{h_j^l} \odot \mathbf{m}_{l,j}(\cdot) = D^l \mathbf{m}_j(\cdot)(1, ..., 1)$,

$$D^l \hat{\mathbf{m}}_j(x_j) \ominus D^l \mathbf{m}_j(x_j) = O_p(n^{-(r+1-l)/(2r+3)})$$

Same error rates with locally polynomial regression for a real-valued response on a single covariate.

Asymptotic distributions

Extend the notion to the tuple of outer products, now operated on $\mathbb{H}^{r+1}$. Define $\mathbf{u} \otimes_{\mathsf{tp}} \mathbf{v} \colon \mathbb{H}^{r+1} \to \mathbb{H}^{r+1}$ by

$$(\mathbf{u} \otimes_{\mathsf{tp}} \mathbf{v})(\mathbf{b}^{\mathsf{tp}}) = ((\mathbf{u} \otimes \mathbf{v})(\mathbf{b}_0), \ldots, (\mathbf{u} \otimes \mathbf{v})(\mathbf{b}_r))^{\top}, \mathbf{b}^{\mathsf{tp}} = (\mathbf{b}_0, \ldots, \mathbf{b}_r)^{\top} \in \mathbb{H}^{r+1}.$$

$\otimes_{\mathsf{tp}}$ is still a linear operator. Now, define the following matrix involving complete moments of K^2:

$$\mathbf{V} = \int_{-1}^{1} \mathsf{vec}(v) \cdot \mathsf{vec}(v)^{\top} K^2(v)\, dv.$$

Recall the definition of the response error, $\varepsilon = Y \ominus E(Y|X)$ and note that $\mathbf{N}$ is invertible since $K(v) > 0$ on an interval with non-empty interior.

Asymptotic distributions

For each $x_j \in (0,1)$ and $1 \le j \le d$, define $\Sigma_j(x_j)\colon \mathbb{H}^{r+1} \to \mathbb{H}^{r+1}$ by

$$\Sigma_j(x_j)(\mathbf{b}) = \alpha_j^{-1} p_j(x_j)^{-1} \cdot \mathbf{N}^{-1}\mathbf{V}\mathbf{N} \odot E[(\varepsilon \otimes_{\mathsf{tp}} \varepsilon)(\mathbf{b})|X_j = x_j], \quad \mathbf{b} \in \mathbb{H}^{r+1}$$

where α_j are the constants in the assumption (A4).

Let $\mathbf{Z}_{0j}(x_j) \overset{d}{=} \mathbf{G}(\mathbf{0}_{r+1}, \boldsymbol{\Sigma}_j(x_j))$ be a Gaussian random element taking values in $\mathbb{H}^{r+1}$ with mean $\mathbf{0}_{r+1}$ and covariance operator $\boldsymbol{\Sigma}_j(x_j)$. It means that, for any $\mathbf{b} \in \mathbb{H}^{r+1}$, $\langle \mathbf{Z}_{0j}(x_j), \mathbf{b} \rangle_{\mathsf{tp}} \sim N(0, \langle \boldsymbol{\Sigma}_j(x_j)(\mathbf{b}), \mathbf{b} \rangle_{\mathsf{tp}})$.

Now, we define $\mathbb{H}^{r+1}$-valued random elements $\mathbf{Z}_j(x_j)$ by

$$\mathbf{Z}_j(x_j) \overset{d}{=} \frac{\alpha_j^{r+1}}{(r+1)!} \mathbf{N}^{-1} \boldsymbol{\gamma} \odot D^{r+1}\mathbf{m}_j(x_j)(1, \ldots, 1) \oplus \mathbf{Z}_{0j}(x_j).$$

and let $\mathbf{Z}(x)^{\mathsf{T}} = \left(\mathbf{Z}_1(x_1)^{\mathsf{T}} \cdots \mathbf{Z}_d(x_d)^{\mathsf{T}} \right)$.

Asymptotic distributions

Let

$$\mathbf{Z}_{n,j}(x_j) = n^{(r+1)/(2r+3)} \odot \left(\hat{\mathbf{m}}_j^{tp}(x_j) \ominus \mathbf{m}_j(x_j) \right), \quad 1 \leq j \leq d.$$

and $\mathbf{Z}_n(x)^{\top} = \left(\mathbf{Z}_{n,1}(x_1)^{\top} \cdots \mathbf{Z}_{n,d}(x_d)^{\top} \right)$.

Theorem 3. Assume that the conditions of Theorem 2 hold. Also, for all $1 \leq j \neq k \leq d$, assume that $E[\|\varepsilon\|^{\alpha}|X_j = \cdot]$ for some $\alpha > (2r+3)/(r+1)$ and $E[\varepsilon \otimes_{\mathsf{tp}} \varepsilon | X_j = \cdot, X_k = \cdot]$ are bounded on $[0,1]$ and $[0,1]^2$, respectively, and that $E(\varepsilon \otimes_{\mathsf{tp}} \varepsilon | X_j = \cdot)$ are continuous on $[0,1]$. Then, for each $\mathbf{x} = (x_1, \ldots, x_d)^{\top} \in [0,1]^d$ it holds that $\mathbf{Z}_n(\mathbf{x})$ converges in distribution to $\mathbf{Z}(\mathbf{x})$.

Asymptotic distributions

Recall

$$\hat{\mathbf{m}}_j^{tp}(x_j) \ominus \mathbf{m}_j^{tp}(x_j) = \tilde{\mathbf{m}}_j^{A,\mathsf{tp}}(x_j) \oplus \frac{h^{r+1}}{(r+1)!}\mathbf{N}_j(x_j)\boldsymbol{\gamma}_j(x_j) \odot D^{r+1}\mathbf{m}_j(x_j)(1, ..., 1)$$

$$\oplus o_p\left(n^{-(r+1)/(2r+3)}\right)$$

Let $\mathbf{S}_n(\mathbf{x})^T = (\mathbf{S}_{n,1}(x_1)^T, .., , \mathbf{S}_{n,d}(x_d)^T$ where $\mathbf{S}_{n,j}(x_j) = \tau_n^{-1} \odot \tilde{\mathbf{m}}_j^{A,tp}(x_j)$.

We show that $\mathbf{S}_n(\mathbf{x})$ converges in distribution to $\mathbf{Z}_0(\mathbf{x}) = \left(\mathbf{Z}_{01}(x_1)^\mathsf{T} \cdots \mathbf{Z}_{0d}(x_d)^\mathsf{T}\right)^T$

Asymptotic distributions

Recall

$$\mathbf{S}_{n,j}(x_j) = n^{-1}\tau_n^{-1}\hat{\mathbf{M}}_{jj}(x_j)^{-1} \odot \bigoplus_{i=1}^{n} \mathrm{vec}\left(\frac{X_{ij}-x_j}{h_j}\right) K_{h_j}(X_{ij}, x_j) \odot \varepsilon_i.$$

Let $\mathbf{S}_n^*(\mathbf{x}) = (\mathbf{S}_{n,1}^*(x_1)^\top, \ldots, \mathbf{S}_{n,d}^*(x_d)^\top)^\top$ with

$$\mathbf{S}_{n,j}^*(x_j) = n^{-1}\tau_n^{-1} p_j(x_j)^{-1}\mathbf{N}^{-1} \odot \bigoplus_{i=1}^{n} \mathrm{vec}\left(\frac{X_{ij}-x_j}{h_j}\right) K_{h_j}(X_{ij}, x_j) \odot \varepsilon_i.$$

Then $\mathbf{S}_n(\mathbf{x}) \ominus \mathbf{S}_n^*(\mathbf{x}) = o_p(1)$ for each fixed $\mathbf{x} \in (0,1)^d$. Below, we apply Theorem 2.3 to prove that $\mathbf{S}_n^*(\mathbf{x}) \xrightarrow{d} \mathbf{Z}_0(\mathbf{x})$.

Asymptotic distributions

Theorem 2.3. For the row-wise sum X_n of a triangular array $\{\varepsilon_{ni}\}$ satisfying (2.1), let $\mathcal{C}_n$ be the covariance operator of X_n. Assume

(i) $\langle \mathcal{C}_n(e_j), e_k \rangle \to c_{jk}$ for all $j, k \geq 1$;

(ii) $\sum_{j=1}^{\infty} \langle \mathcal{C}_n(e_j), e_j \rangle \to \sum_{j=1}^{\infty} c_{jj} < \infty$;

(iii) for all $\delta > 0$ and $j \geq 1$,

$$\sum_{i=1}^{n} \mathbb{E}\Big(\langle \varepsilon_{ni}, e_j \rangle^2 I\big(|\langle \varepsilon_{ni}, e_j \rangle| > \delta \big) \Big) \to 0.$$

Then, $X_n \rightsquigarrow G(0, \mathcal{C})$, where the covariance operator $\mathcal{C}$ is characterized by

$$\langle \mathcal{C}(x), e_j \rangle = \sum_{k=1}^{\infty} c_{jk} \cdot \langle x, e_k \rangle, \quad j \geq 1. \quad \square$$

Asymptotic distributions

For the first condition,

$$\langle \mathcal{C}_n(\mathbf{e}_{s,j}^{\mathrm{F,tp}}), \mathbf{e}_{s',k}^{\mathrm{F,tp}} \rangle = E\left(\left\langle \mathbf{S}_n^*(\mathbf{x}), \mathbf{e}_{s,j}^{\mathrm{F,tp}} \right\rangle_{\mathbb{H}^{(r+1)d}} \cdot \left\langle \mathbf{S}_n^*(\mathbf{x}), \mathbf{e}_{s',k}^{\mathrm{F,tp}} \right\rangle_{\mathbb{H}^{(r+1)d}} \right)$$

$$= n^{-1}\tau_n^{-2} p_j(x_j)^{-1} p_k(x_k)^{-1} E\left[\left\langle K_{h_j}(X_{ij}, x_j) \cdot \mathbf{N}^{-1} \operatorname{vec}\left(\frac{X_{ij} - x_j}{h_j} \right) \odot \varepsilon_i, \mathbf{e}_s^{\mathrm{tp}} \right\rangle_{\mathbb{H}^{r+1}} \right.$$

$$\left. \cdot \left\langle K_{h_k}(X_{ik}, X_{ik}) \cdot \mathbf{N}^{-1} \operatorname{vec}\left(\frac{X_{ik} - x_k}{h_k} \right) \odot \varepsilon_i, \mathbf{e}_{s'}^{\mathrm{tp}} \right\rangle_{\mathbb{H}^{r+1}} \right].$$

For doing this, we find that the following identity is useful.

$$\langle (\mathbf{u} \odot \varepsilon, \mathbf{b}_1) \rangle_{\mathbb{H}^{r+1}} \cdot \langle (\mathbf{v} \odot \varepsilon, \mathbf{b}_2) \rangle_{\mathbb{H}^{r+1}} = \left\langle \mathbf{v}\mathbf{u}^\top \odot (\varepsilon \otimes_{\mathrm{tp}} \varepsilon)(\mathbf{b}_1), \mathbf{b}_2 \right\rangle_{\mathbb{H}^{r+1}},$$

$$\mathbf{u}, \mathbf{v} \in \mathbb{R}^{r+1}, \mathbf{b}_1, \mathbf{b}_2 \in \mathbb{H}^{r+1}.$$

Asymptotic distributions

If $j = k$,

$$\langle \mathcal{C}_n(\mathbf{e}_{s,j}^{\mathrm{F,tp}}), \mathbf{e}_{s',k}^{\mathrm{F,tp}} \rangle = n^{-1}\tau_n^{-2} p_j(x_j)^{-2} E\left[K_{h_j}(X_{ij}, x_j)^2 \left\langle \mathbf{N}^{-1}\,\mathrm{vec}\left(\frac{X_{ij}-x_j}{h_j}\right), \mathbf{e}_s^{\mathrm{tp}} \right\rangle \right.$$

$$\left. \cdot \left\langle \mathrm{vec}\left(\frac{X_{ij}-x_j}{h_j}\right)^\top \mathbf{N}^{-1} \odot (\varepsilon_i \otimes_{\mathrm{tp}} \varepsilon_i)(\mathbf{e}_s^{\mathrm{tp}}), \mathbf{e}_{s'}^{\mathrm{tp}} \right\rangle \right]_{\mathbb{H}^{r+1}}$$

$$= n^{-1}\tau_n^{-2} h_j^{-1} p_j(x_j)^{-1} \langle \mathbf{N}^{-1}\mathbf{V}\mathbf{N}^{-1} \odot E\left[(\varepsilon \otimes_{\mathrm{tp}} \varepsilon)(\mathbf{e}_s^{tp}) \,|X_j = x_j \right], \mathbf{e}_{s'}^{\mathrm{tp}} \rangle_{\mathbb{H}^{r+1}} (1 + o(1))$$

$$= \left\langle \mathbf{\Sigma}_j(x_j)(\mathbf{e}_s^{\mathrm{tp}}), \mathbf{e}_{s'}^{\mathrm{tp}} \right\rangle_{\mathbb{H}^{r+1}} \cdot (1 + o_p(1))$$

Asymptotic distributions

If $j \neq k$, using

$$\sup_{(u,v)\in[0,1]^2} \| E\left[(\varepsilon \otimes_{\text{tp}} \varepsilon)(\mathbf{b}) \,|\, X_j = u, X_k = v \right] \|_{\mathbb{H}^{r+1}} \leq C \cdot \|\mathbf{b}\|_{\mathbb{H}^{r+1}}$$

We get

$$\begin{aligned} \langle \mathcal{C}_n(\mathbf{e}^{\text{F,tp}}_{s,j}), \mathbf{e}^{\text{F,tp}}_{s',k} \rangle &= n^{-1}\tau_n^{-2} p_j(x_j)^{-1} p_k(x_k)^{-1} \Big| E\Big[K_{h_j}(x_j, X_{ij}) K_{h_k}(x_k, X_{ik}) \\ &\cdot \left\langle \mathbf{N}^{-1} \operatorname{vec}\left(\frac{X_{ij}-x_j}{h_j}\right) \operatorname{vec}\left(\frac{X_{ik}-x_k}{h_k}\right)^{\top} \mathbf{N}^{-1} \ \odot (\varepsilon_i \otimes_{\text{tp}} \varepsilon_i)(\mathbf{e}^{\text{tp}}_s), \mathbf{e}^{\text{tp}}_{s'} \right\rangle_{\mathbb{H}^{r+1}} \Big] \Big| \\ &\leq C \cdot n^{-1}\tau_n^{-2} p_j(x_j)^{-1} p_k(x_k)^{-1} \Big\| \int_{[0,1]^2} K_{h_j}(x_j, u) K_{h_k}(x_k, v) \\ &\cdot \mathbf{N}^{-1} \operatorname{vec}\left(\frac{u-x_j}{h_j}\right) \operatorname{vec}\left(\frac{v-x_k}{h_k}\right)^{\top} \mathbf{N}^{-1} \cdot p_{jk}(u,v)\, du\, dv \Big\|_{\text{op}} \\ &= C \cdot n^{-1/(2r+3)} \cdot \frac{p_{jk}(x_j,x_k)}{p_j(x_j)p_k(x_k)} \|\mathbf{N}^{-1}\boldsymbol{\mu}\boldsymbol{\mu}^{\top}\mathbf{N}^{-1}\|_{\text{op}} \cdot (1+o(1)) \to 0. \end{aligned}$$

$$\langle \boldsymbol{\Sigma}(\mathbf{x})(\mathbf{e}_{s,j}^{\mathrm{F,tp}}), \mathbf{e}_{s',k}^{\mathrm{F,tp}} \rangle_{\mathbb{H}^{(r+1)d}} = \langle \boldsymbol{\Sigma}_j(x_j)(\mathbf{e}_s^{\mathrm{tp}}), \mathbf{e}_{s'}^{\mathrm{tp}} \rangle_{\mathbb{H}^{r+1}} \cdot I(k=j)$$

where $\boldsymbol{\Sigma}(\mathbf{x})(\mathbf{a}) = \left(\boldsymbol{\Sigma}_1(x_1)(\mathcal{P}_1 \odot \mathbf{a})^T, ..., \boldsymbol{\Sigma}_d(x_d)(\mathcal{P}_d \odot \mathbf{a})^T\right)^T$
$\mathcal{P}_j = (\mathbf{0}_{(r+1)\times(j-1)(r+1)}, \mathbf{I}_{(r+1)}, \mathbf{0}_{(r+1)\times(d-j)(r+1)})$
So we get

$$\langle \mathcal{C}_n(\mathbf{e}_{s,j}^{\mathrm{F,tp}}), \mathbf{e}_{s',k}^{\mathrm{F,tp}} \rangle \to \langle \boldsymbol{\Sigma}(\mathbf{x})(\mathbf{e}_{s,j}^{\mathrm{F,tp}}), \mathbf{e}_{s',k}^{\mathrm{F,tp}} \rangle$$

Asymptotic distributions

Second condition

$$\sum_{s=1}^{\infty}\sum_{j=1}^{d}\langle \mathcal{C}_n(\mathbf{e}_{s,j}^{\mathrm{F,tp}}), \mathbf{e}_{s,j}^{\mathrm{F,tp}}\rangle_{\mathbb{H}^{(r+1)d}} = \sum_{s=1}^{\infty}\sum_{j=1}^{d} E\left(\left\langle \mathbf{S}_n^*(\mathbf{x}), \mathbf{e}_{s,j}^{F,\mathrm{tp}}\right\rangle_{\mathbb{H}^{(r+1)d}}^2\right)$$

$$= \sum_{s=1}^{\infty}\sum_{j=1}^{d} \alpha_j^{-1} p_j(x_j)^{-1} \langle \mathbf{N}^{-1}\mathbf{V}\mathbf{N}^{-1} \odot E[(\boldsymbol{\varepsilon} \otimes_{\mathrm{tp}} \boldsymbol{\varepsilon})(\mathbf{e}_s^{\mathrm{tp}})|X_j = x_j], \mathbf{e}_s^{\mathrm{tp}}\rangle_{\mathbb{H}^{r+1}}$$

$$\leq \lambda_{\max}(\mathbf{N}^{-1}\mathbf{V}\mathbf{N}^{-1}) \cdot \sum_{s=1}^{\infty}\sum_{j=1}^{d} \alpha_j^{-1} p_j(x_j)^{-1} \langle E[(\boldsymbol{\varepsilon} \otimes_{\mathrm{tp}} \boldsymbol{\varepsilon})(\mathbf{e}_s^{\mathrm{tp}}) \mid X_j = x_j], \mathbf{e}_s^{\mathrm{tp}}\rangle_{\mathbb{H}^{r+1}}$$

$$= (r+1) \cdot \lambda_{\max}(\mathbf{N}^{-1}\mathbf{V}\mathbf{N}^{-1}) \cdot \sum_{j=1}^{d} \alpha_j^{-1} p_j(x_j)^{-1} \cdot E(\|\boldsymbol{\varepsilon}\|^2 \mid X_j = x_j)$$

$$< \infty.$$

Asymptotic distributions

Now, for the third condition we write $\mathbf{S}_n^*(\mathbf{x}) = \bigoplus_{i=1}^n \mathbf{S}_n^{*(i)}(\mathbf{x})$. Then, we may prove

$$\sum_{i=1}^n E\|\mathbf{S}_n^{*(i)}(\mathbf{x})\|_{\mathbb{H}^{(r+1)d}}^{2+\delta} \leq C' n^{-((r+1)/(2r+3))\delta}$$

for some constant $0 < C' < \infty$, where $0 < \delta < \alpha - 2$ and α is the constant in the condition of Theorem 3.